

Optimize Linear Antenna Arrays with Dandelion Optimization Algorithm

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Abstract—This paper introduces the application of dandelion optimization (DO) algorithm in linear antenna arrays. This is the first time that DO has been used in the optimization of antenna arrays. DO is a new type of evolutionary algorithm inspired by nature, inspired by dandelions' wind-borne behavior in nature. The algorithm iteratively updates the population from three stages: ascending, descending and landing, and finally finds the optimal position. DO can be used to optimize linear antenna arrays to get the antenna pattern geometry we need. In this work, we use the following methods to conduct research experiments: by controlling the antenna array element spacing and phase unchanged to optimize the current amplitude for an array pattern with minimum side lobe level (SLL). To demonstrate how DO can be used for linear array optimization, two examples are given. In terms of suppressing peak SLL in comparison with nonoptimized and other known metaheuristic optimization techniques. The optimization ability of DO is better than them. This demonstrates the feasibility of the DO algorithm for optimizing antenna arrays and even other problems in electromagnetism.

Keywords—Dandelion optimizer; antenna array; optimization; algorithm; antenna current amplitude

I. INTRODUCTION

Nowadays, there are more and more evolutionary algorithms inspired by nature, and they are becoming more and more popular in research and development in many various fields. They have been shown by many scholars to be very effective in solving very difficult multidimensional and non-convex problems in engineering. More and more natural heuristic algorithms have been discovered, and some are also very famous, for example: ant colony optimization (ACO) [1], particle swarm optimization (PSO) [2], genetic algorithm (GA) [3], etc. In general, each algorithm has its own advantages and disadvantages, and there is no perfect algorithm in the world that happens to solve all optimization problems. Hence, discovering and studying a new algorithm or innovating on the basis of the existing algorithm to better optimize performance has become a hot topic for some researchers recently.

As we all know, the design of antenna arrays is a nonlinear multidimensional problem in electromagnetics, which is usually very difficult if conventional optimization techniques are used in general. Antenna arrays is a special type of antenna

consisting of not less than two antenna elements that are regularly or randomly arranged and obtain predetermined radiation characteristics by appropriate excitation. For linear array, keep the width of the main lobe constant, the lower the side lobe, the deeper the zero point, the more concentrated the energy it radiates, the better the performance of the antenna. Hence, a number of nature-inspired optimization strategies have been used. For example, the ACO [4], spider monkey optimization [5] and cat swarm optimization [6] have also been successfully applied for optimization of linear arrays. Therefore, the optimization algorithm of natural heuristics has exerted an important influence in the field of electromagnetism and has been widely used.

However, this is the first time of DO being used for antenna array optimization. In this article, DO will be applied to the line array using the following optimization method to obtain an optimized pattern synthesis: by optimizing the antenna current amplitudes while assuming uniform space and phase excitation. The algorithm is described in detail in Section 2. Then, in Section 3, two examples of line array synthesis are presented and DO-optimized antenna currents are given along with corresponding array patterns. By comparing with other natural evolution algorithms, the results obtained were verified. Finally, a summary is made in Section 4.

II. DANDELION OPTIMIZATION ALGORITHM

This section describes dandelion optimization algorithm in detail, including mathematical models and expressions for the DO algorithm.

The following section begins with an introduction to the mathematical model of the dandelion algorithm.

A. Mathematical Model

The mathematical model for constructing dandelion algorithm mainly includes four stages, namely population initialization, ascending phase, descent phase and landing phase.

1) Initialization

First at initialization time, its population number can be expressed using the following formula

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$$\text{population} = \begin{bmatrix} x_1^1 & \dots & x_1^{Dim} \\ \dots & \dots & \dots \\ x_{pop}^1 & \dots & x_{pop}^{Dim} \end{bmatrix} \quad (1)$$

Where pop denotes the population size and Dim is the dimension of the variable [7]. UB represents the upper bound and LB represents the lower bound. Each solution is randomly distributed within the upper and lower bounds, and the expression of the i_{th} individual X_i is

$$X_i = \text{rand} \times (UB - LB) + LB \quad (2)$$

Where i is an integer between 1 and pop and rand denotes a random number between 0 and 1. LB and UB are expressed as

$$\begin{aligned} LB &= [lb_1, \dots, lb_{Dim}] \\ UB &= [ub_1, \dots, ub_{Dim}] \end{aligned} \quad (3)$$

This article takes the minimum value as an example, and, the mathematical expression X_{elite} of the initial optimal body is

$$\begin{aligned} f_{best} &= \min(f(X_i)) \\ X_{elite} &= X(\text{find}(f_{best} = f(X_i))) \end{aligned} \quad (4)$$

Where $\text{find}()$ denotes two indexes with equal values.

2) Ascending phase

In this paper, the weather is divided into sunny and rainy days, which analyze the dispersal of dandelion seeds separately.

The first case: On a sunny day, wind speeds can be regarded to have a lognormal distribution $\ln Y \sim N(\mu, \sigma^2)$. Such a distribution allows dandelion seeds to fly farther away. Under the action of the wind, the dandelion spirals upwards. In this case, the mathematical expression for the dandelion evolutionary iteration is

$$X_{t+1} = X_t + \alpha * v_x * v_y * \ln Y * (X_s - X_t) \quad (5)$$

Where X_t represents the position of the dandelion seed at the t_{th} iteration. X_s represents the randomly selected position in the search space during iteration t . Eq. (6) provides the expression for the randomly generated position.

$$X_s = \text{rand}(1, Dim) * (UB - LB) + LB \quad (6)$$

$\ln Y$ denotes a lognormal distribution subject to $\mu = 0$ and $\sigma^2 = 1$, and its mathematical formula is

$$\ln Y = \begin{cases} \frac{1}{y\sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma^2}(\ln y)^2\right], & y \geq 0 \\ 0, & y < 0 \end{cases} \quad (7)$$

Eq. (7), α is an adaptive parameter used to adjust the search step length, and the mathematical formula is

$$\alpha = \text{rand}() * \left(\frac{1}{T^2}t^2 - \frac{2}{T}t + 1\right) \quad (8)$$

Eq. (8), α is a random number between $[0, 1]$ and follows the principle of nonlinear reduction. Dandelions produce vortices during ascent, while V_x and V_y represent the two components of the force generated by the vortex. To calculate V_x and V_y , we need to use (9).

$$\begin{aligned} r &= \frac{1}{e^\theta} \\ v_x &= r * \cos \theta \\ v_y &= r * \sin \theta \end{aligned} \quad (9)$$

Where θ is a random number between $[-\pi, \pi]$.

The second case: In case of rain, dandelion seeds can only be exploited in a small neighborhood near them, and the mathematical expression for this stage is

$$X_{t+1} = X_t * k \quad (10)$$

Where k is used to specify the area of dandelion during local search, and (11) is used to calculate the domain.

$$\begin{aligned} q &= \frac{1}{T^2 - 2T + 1}t^2 - \frac{2}{T^2 - 2T + 1}t + 1 + \frac{1}{T^2 - 2T + 1} \\ k &= 1 - \text{rand}() * q \end{aligned} \quad (11)$$

For both weather conditions mentioned above, The evolutionary iteration mathematical model of dandelion is

$$X_{t+1} = \begin{cases} X_t + \alpha * v_x * v_y * \ln Y * (X_s - X_t), & \text{rand} < 1.5 \\ X_t * k, & \text{else} \end{cases} \quad (12)$$

The above section describes the iterative mathematical model of dandelion under different weather conditions during the ascent phase and its flight propagation behavior.

3) Descent phase

In this stage, after the dandelion seeds have risen to a certain height, they begin to decline steadily. The mathematical expression corresponding to this stage is

$$X_{t+1} = X_t - \alpha * \beta_t * (X_{\text{mean}_t} - \alpha * \beta_t * X_t) \quad (13)$$

Where β_t denotes Brownian motion, it is a random number that obeys a standard normal distribution. X_{mean_t} denotes the average position of the population in the i_{th} iteration, and its mathematical formula is

$$X_{\text{mean}_t} = \frac{1}{pop} \sum_{i=1}^{pop} X_i \quad (14)$$

This section describes the iterative optimization mathematical model of dandelion in the descent phase.

4) Landing phase

In this part, dandelion seeds are chosen to land randomly. As the iteration progresses, the algorithm converges to the global optimal solution. This behavior is expressed in (15).

$$X_{t+1} = X_{elite} + \text{levy}(\lambda) * \alpha * (X_{elite} - X_t * \delta) \quad (15)$$

Where X_{elite} represents the optimal position of the dandelion seed in the i_{th} iteration. $\text{levy}(\lambda)$ represents the function of Levy flight and its calculation formula is

$$\text{Levy}(\lambda) = s * \frac{w * \sigma}{|t|^{\frac{1}{\beta}}} \quad (16)$$

Eq. (16), β take the value of 1.5 in this paper. s is a fixed value of 0.01. w and t are random numbers between $[0, 1]$. The mathematical formula of σ is

$$\sigma = \left(\frac{\Gamma(1+\beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \right) \quad (17)$$

Where β is fixed at 1.5. δ is a function that increases linearly from $[0, 2]$ and its expression is expressed by (18).

$$\delta = \frac{2t}{T} \quad (18)$$

This part describes the process of dandelion during the landing phase.

III. LINEAR ANTENNA ARRAY SYNTHESIS

A. Geometry of Linear Antenna Array

As shown, a geometric representation of a linear antenna array in Fig. 1.

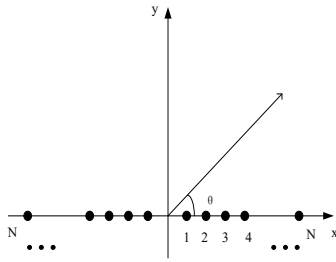


Figure. 1 The geometric representation of antenna array.

Due to symmetry, the array factor (AF) in the azimuth plane is given by

$$AF(\theta) = 2 \sum_{n=1}^N I_n \cos(kx_n \cos(\theta) + \psi_n) \quad (19)$$

Where I_n , ψ_n , and X_n are the excitation amplitude, phase, and position of n th element in the array. k is the wave number and is given by $2\pi/\lambda$ and θ is the azimuth angle.

The array factor for antenna current amplitude optimization is given by

$$AF(\theta) = 2 \sum_{n=1}^N I_n \cos(kx_n \cos(\theta)) \quad (20)$$

Uniform phase, that is, $\psi_n = 0$, and the element spacing (X_n) is $\lambda/2$.

B. Antenna Current Optimization

This section, examples 1–2 are presented for optimization of the antenna current amplitudes (I_n) for optimal pattern synthesis. Let the phase excitation be uniform, that is, $\psi_n = 0$. Element spacing is set to $\lambda/2$. The element spacing of the traditional uniform array is also $\lambda/2$.

1) Reduce the peak of side lobes

The mathematical expression used to describe the fitness function to optimize the peak SLL to the minimum is

$$\text{Fitness} = \min \left(\max \left(20 \log |AF(\theta)| \right) \right) \quad (21)$$

Here, $\max \left(20 \log |AF(\theta)| \right)$ gives the maximum SLL (peak SLL). $AF(\theta)$ is the array factor given by (20). The DO runs independently 20 times on MATLAB with each iteration set to 1000 and population(N) set to 40.

TABLE I. OPTIMIZED CURRENT AMPLITUDES FOR PEAK SLL MINIMIZATION

Linear array type	Method	Optimized amplitudes	current
16 elements	PSO[8]	1.0000,0.9521,0.5605,0.7372	
		0.5940,0.4465,0.3079,0.2724	
	ALO[9]	1.0000,0.9344,0.8521,0.7044	
		0.6000,0.4000,0.3003,0.2002	
	DO	0.9452,0.9046,0.7913,0.6818	
		0.5261,0.3825,0.2597,0.1767	

Design Example 1 presents the design of $2N = 16$ element linear array for achieving the lowest side lobe in the regions, $\theta = [0^\circ, 76^\circ]$ and $[104^\circ, 180^\circ]$. The amplitudes of the optimized current obtained by using the proposed method are shown in Table 1 and the pattern of the array is shown in Fig. 2. DO gives peak SLL of -33.37 dB which are 20.22 dB and 2.74 dB lower as compared to uniform array and as compared to PSO [8] optimized array. The peak SLL has been lowered from -30.85 dB to -33.37 dB (by 2.52 dB) as compared to ALO [9] optimized array. These results are depicted in Fig. 2 and summarized in Table 3.

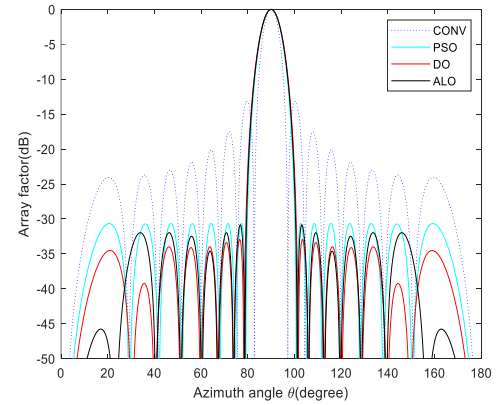


Figure. 2 Array pattern for design Example 1.

Design Example 2 presents the design of $2N = 24$ element linear array for achieving minimum SLL. In this case, DO is applied to prove its efficacy. The array pattern is given in Figs. 3. In table 2, the optimized current amplitude values optimized using DO and other algorithms are given. We can clearly get it from the results. MSMO [10] provides a peak SLL of -37.52 dB and RRA [11] provides a peak SLL of -41.08 dB. DO gives peak SLL of -42.56 dB indicating a substantial gain from its nearest competitor. These results which have been summarized in Table 3.

TABLE II. OPTIMIZED EXCITATION AMPLITUDES OF $2N = 24$ ELEMENTS

Method	Optimized current amplitudes	PSL (dB)
CONV(Uniform array)	1.0000,1.0000,1.0000,1.0000,1.0000,1.0000 1.0000,1.0000,1.0000,1.0000,1.0000,1.0000	-13.23
MSMO[10]	1.0000,0.9717,0.9195,0.8438,0.7555,0.6565 0.5278,0.4534,0.3194,0.2430,0.1818,0.1296	-37.52
RRA[11]	1.0000,0.9706,0.9141,0.8344,0.7371,0.6287 0.5162,0.4060,0.3038,0.2140,0.1395,0.1136	-41.08
DO	0.8192,0.7901,0.7414,0.6798,0.5855,0.5004 0.4069,0.3111,0.2287,0.1596,0.1018,0.0692	-42.56

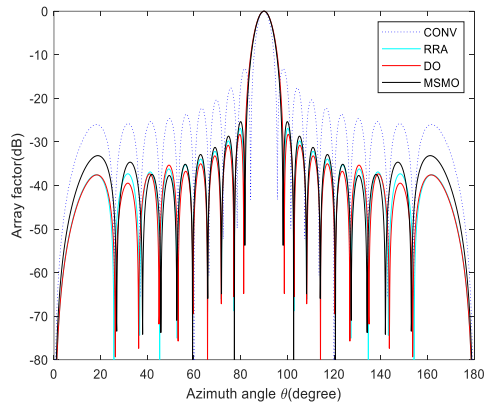


Figure. 3 Array pattern for design example 2.

TABLE III. BENCHMARKING RESULTS FOR DESIGN EXAMPLE 1 AND 2 FOR PEAK SLL SUPPRESSION

	Peak SLL (dB)	
	Example 1	Example 2
Uniform array	-13.15	-13.23
PSO[8]	-30.63	~
ALO[9]	-30.85	~
MSMO[10]	~	-37.52
RRA[11]	~	-41.08
DO	-33.37	-42.56

IV. CONCLUSION

This paper briefly introduces the principle and mathematical model of dandelion algorithm, and applies it to the optimization application of linear antenna array for the first time. DO was used to optimize the current amplitude of the antenna to obtain the antenna pattern of the smallest side lobe. The simulation results show that compared with other existing algorithms, the DO algorithm performs better in suppressing the peak SLL problem. This demonstrates the feasibility of the DO algorithm for optimizing antenna arrays and even other problems in electromagnetism.

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